

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \quad x > 0, y > 0$$

Final Graphical Interface

Beta Function Calculator

Compute B(x, y) for finite positive real inputs.

Inputs

x (Alt+X)abc

y (Alt+Y)1

Calculate

Swap x and y

Clear

Result and status

Input could not be accepted.

x must be a real number, such as 2.5 or 1e-3.

Calculation history

x	y	B(x, y)
1	1	1
3	2	0.0833333333333
2	3	0.0833333333333

Enter: calculate Esc: clear Alt+X / Alt+Y: focus input

The live interface shows a field-specific nonnumeric-input message while preserving earlier successful calculations. The user can correct the input and continue; native Tkinter widgets retain familiar focus and tab behavior.

Accessibility and Recovery

Keyboard access
Enter, Esc, Alt+X, Alt+Y

Recognition over recall
visible labels and actions

Helpful recovery
field-specific text messages

Status is not color-only
words identify success or error

The GUI aims to be accessible through readable labels, visible focus, redundant text feedback, key-board operation, and recoverable validation. These choices align the interface with the user’s task model and established usability heuristics (Kamthan, 2026c; Nielsen Norman Group, 2026).

Exception policy: expected input and numeric-range errors receive specific messages; unexpected arithmetic failures are not hidden behind a generic message and remain available for diagnosis.

D3 conclusion: the release improves numerical stability, accessibility, and software quality through a stable log-beta implementation, explicit range behavior, and repeatable verification.

Problem 7: Quality and Versioning

Flake8 passed: no violations found.

Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

Correction made:

Pylint reported too-many-public-methods. Splitting one oversized test class into three focused classes restored 10.00/10 without suppressing the rule.

Why 1.0.0:

the first stable GUI and numerical contract, recorded by repository tag v1.0.0 (Semantic Versioning, 2026).

Problem 7: Debugger Evidence

```
(Pdb) break beta_core.beta_function
Breakpoint 1 at /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/beta_core.py:256
(Pdb) continue
(Pdb) > /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/beta_core.py(263)beta_function()
-> if not is_finite(x_value):
(Pdb) p (x_value, y_value)
(2.0, 3.0)
(Pdb) next
> /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/beta_core.py(265)beta_function()
-> if not is_finite(y_value):
(Pdb) next
> /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/beta_core.py(267)beta_function()
-> if x_value <= 0.0:
(Pdb) next
> /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/beta_core.py(269)beta_function()
-> if y_value <= 0.0:
(Pdb) continue
B(2, 3) = 0.083333333333313
The program finished and will be restarted
> /Users/weihuang/Documents/SOEN_6011/D3_F6_Beta_Function/debug_beta.py(1)<module>()
-> """Small driver used only to demonstrate pdb on the Beta calculation."""
(Pdb) █
```

At a breakpoint in beta_function, pdb inspects (x,y) = (2,3) and steps through the finite and domain checks before confirming B(2,3) = 1/12.

Verified Numerical Contract

External verification only – mpmath, 100 decimal digits

Seed: 6011; generated cases: 500

Inputs: log-uniform from 1e-12 through 1e12

Normal representable results checked: 414

Range-limited reference results excluded: 86

Failures above 1e-10 relative error: 0

Maximum relative error: 1.223e-11

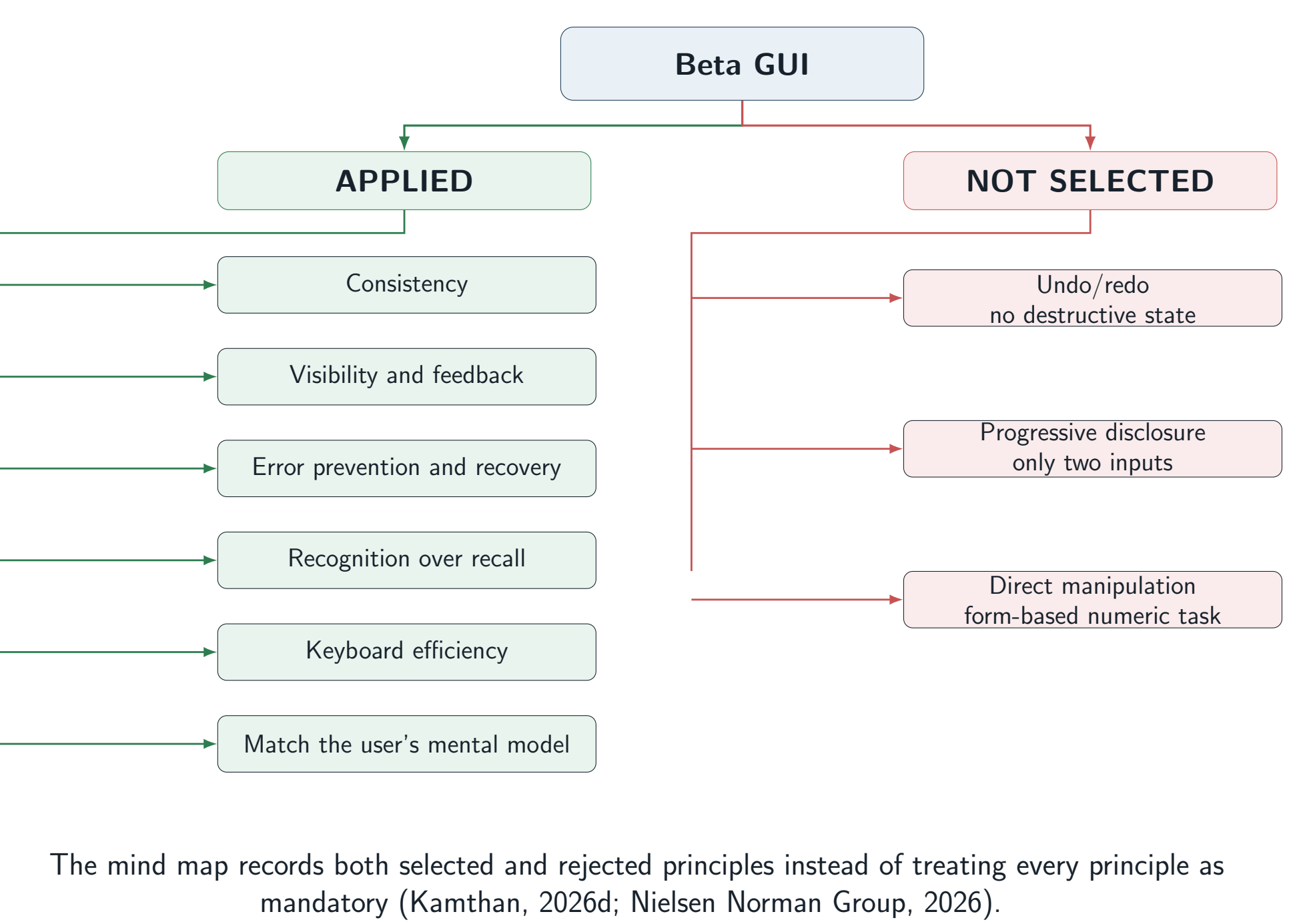
Worst normal-result input: (131117.64197605842, 81.32262207870075)

Large regression B(1e100, 1) relative error: 5.419e-13

Roundable boundary exp(−745): 4.9406564584124654e-324

Contract: accepted inputs return a nonzero finite binary64 result when representable; otherwise the program reports over-flow or underflow. Normal results in the deterministic oracle set meet 10^{−10} relative error; subnormals follow binary64 spacing (NIST DLMF, 2026; mpmath, 2023).

UIDP Decision Mind Map



Problem 8: PyUnit Test Suite

```
test_half_half_equals_pi (test_beta_core.BetaFunctionValueTests.test_half_half_equals_pi)
B(0.5, 0.5) equals pi. ... ok
test_identity_one_y (test_beta_core.BetaFunctionValueTests.test_identity_one_y)
B(1, y) equals 1/y. ... ok
test_identity_x_one (test_beta_core.BetaFunctionValueTests.test_identity_x_one)
B(x, 1) equals 1/x. ... ok
test_known_value_one_one (test_beta_core.BetaFunctionValueTests.test_known_value_one_one)
B(1, 1) equals one. ... ok
test_known_value_two_three (test_beta_core.BetaFunctionValueTests.test_known_value_two_three)
B(2, 3) equals one twelfth. ... ok
test_near_maximum_asymmetric_identity (test_beta_core.BetaFunctionValueTests.test_near_maximum_asymmetric_identity)
A near-maximum finite x does not trigger log-gamma cancellation. ... ok
test_regression_large_asymmetric_identity (test_beta_core.BetaFunctionValueTests.test_regression_large_asymmetric_identity)
The D3 large-input cancellation defect remains corrected. ... ok
test_symmetry (test_beta_core.BetaFunctionValueTests.test_symmetry)
Swapping x and y does not change the results. ... ok
test_missing_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_missing_input_is_rejected)
The GUI parser identifies a missing field before calculation. ... ok
test_negative_x_is_rejected (test_beta_core.InterfaceAndInputTests.test_negative_x_is_rejected)
A negative x is rejected with a field-specific error. ... ok
test_negative_y_is_rejected (test_beta_core.InterfaceAndInputTests.test_negative_y_is_rejected)
A negative y is rejected with a field-specific error. ... ok
test_nonfinite_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_nonfinite_input_is_rejected)
NaN and positive or negative infinity are rejected. ... ok
test_nonnumeric_input_is_rejected (test_beta_core.InterfaceAndInputTests.test_nonnumeric_input_is_rejected)
The GUI parser reports a useful nonnumeric-input error. ... ok
test_overflow_is_reported (test_beta_core.InterfaceAndInputTests.test_overflow_is_reported)
A result above the largest binary64 value reports a range error. ... ok
test_scientific_notation_parser (test_beta_core.InterfaceAndInputTests.test_scientific_notation_parser)
The GUI parser accepts scientific notation. ... ok
test_underflow_is_reported (test_beta_core.InterfaceAndInputTests.test_underflow_is_reported)
An unrepresentable positive result is not returned as zero. ... ok
test_unexpected_arithmetic_error_is_not_hidden (test_beta_core.InterfaceAndInputTests.test_unexpected_arithmetic_error_is_not_hidden)
Unexpected arithmetic failures remain visible for diagnosis. ... ok
test_zero_is_rejected (test_beta_core.InterfaceAndInputTests.test_zero_is_rejected)
Zero is rejected because it is outside the Beta domain. ... ok
test_zero_y_is_rejected (test_beta_core.InterfaceAndInputTests.test_zero_y_is_rejected)
A zero y is rejected with a field-specific error. ... ok
test_roundable_subnormal_exponential (test_beta_core.NumericalBoundaryTests.test_roundable_subnormal_exponential)
A finite result near the subnormal boundary is not rejected. ... ok
test_unroundable_exponential_is_rejected (test_beta_core.NumericalBoundaryTests.test_unroundable_exponential_is_rejected)
An exponential below the rounding boundary reports underflow. ... ok

Ran 21 tests in 0.002s
OK
```

21/21 tests pass. Coverage includes trusted values, B(1,y) = 1/y, B(x,1) = 1/x, symmetry, the D3 large-input regression, separate x/y domain errors, NaN/infinity, overflow, underflow, and both sides of the subnormal rounding boundary.

Generative AI and CASTROFF. Tool: OpenAI ChatGPT/Codex. **Constraints:** preserve from-scratch code and honest range claims; **Audience:** TA/instructor; **Structure:** evidence by Problems 7 and 8; **Tone:** concise and technical; **Role:** graduate student reviewer; **Output format:** code/test suggestions and poster statements; **Focus:** quality, accessibility, binary64 boundaries, and regression coverage; **Function:** identify missing evidence and unsupported claims. Example prompts requested a PEP 8 and range-behaviour review and focused PyUnit regressions. Suggestions were executed and revised; unverified claims were rejected. Full records are in GAI.USAGE.md.

References. Kamthan, P. (2026a), *SOEN 6011 Project Description*; Kamthan, P. (2026b), *Building, Versioning, and Releasing Software*; Kamthan, P. (2026c), *Introduction to Mental Models*; Kamthan, P. (2026d), *Brainstorming and Mind Mapping*; Nielsen Norman Group (2026), *Ten Usability Heuristics*; Semantic Versioning (2026), *Semantic Versioning 2.0.0*; NIST DLMF (2026), *Section 5.12: Beta Function*; mpmath (2023), *mpmath 1.3.0 Documentation*; Van Rossum, Warsaw, and Coghlan (2001), *PEP 8*; Python Software Foundation (2026), *unittest and pdb Documentation*.

Repository: github.com/HW925/SOEN-6011-D3-Beta-Function